

Data Visualisation & Analytics

100 Practice MCQs

With Answers

Covers: Google Sheets • Python (NumPy, Pandas, Matplotlib, Seaborn) • Tableau

Each question is followed immediately by its correct answer and brief explanation.

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Q1. Which type of analytics answers the question “What happened?”?

- (A) Predictive (B) Prescriptive
(C) Descriptive (D) Diagnostic

Answer

(C) Descriptive Analytics summarises historical data to describe what has happened (e.g., monthly sales reports).

Q2. What is the correct order of the analytics lifecycle?

- (A) Analysis → Collection → Cleaning (B) Collection → Cleaning → EDA →
→ Visualisation Analysis → Visualisation
(C) Cleaning → Collection → Modelling (D) Visualisation → Collection → Clean-
→ Reporting ing → Analysis

Answer

(B) The standard lifecycle is: Problem Definition → Data Collection → Cleaning → EDA → Modelling → Visualisation → Reporting.

Q3. In Google Sheets, what does \$A\$1 represent?

- (A) Relative reference (B) Mixed reference (col fixed)
(C) Absolute reference (D) Mixed reference (row fixed)

Answer

(C) Absolute reference — both the column (\$A) and row (\$1) are locked and do not change when the formula is copied.

Q4. Which shortcut cycles through reference types in Google Sheets?

- (A) Ctrl+R (B) F2
(C) F4 (D) Ctrl+\$

Answer

(C) Pressing F4 cycles through Relative → Absolute → Mixed (row) → Mixed (col) references.

Q5. What function removes leading and trailing spaces from a text string?

- (A) CLEAN() (B) STRIP()
(C) TRIM() (D) REMOVE()

Answer

(C) TRIM(text) removes extra leading, trailing, and internal duplicate spaces from a string.

Q6. Which function returns the position of a search string (case-insensitive)?

- (A) FIND() (B) SEARCH()
(C) LOCATE() (D) POSITION()

Answer

(B) SEARCH() is case-insensitive. FIND() is case-sensitive. Neither exists as LOCATE or POSITION in Sheets.

Q7. What is the correct syntax for VLOOKUP to find an exact match?

- (A) =VLOOKUP(key, range, col, TRUE) (B) =VLOOKUP(range, key, col, FALSE)
(C) =VLOOKUP(key, range, col, FALSE) (D) =VLOOKUP(key, col, range, 0)

Answer

(C) =VLOOKUP(key, range, col_index, FALSE) — FALSE (or 0) means exact match.

Q8. VLOOKUP can look up values to the LEFT of the key column.

- (A) True — always (B) True — with a helper column
(C) False — key must be in the first column of range (D) False — it only searches rows

Answer

(C) VLOOKUP requires the key to be in the first column of the range; it can only return values to the right.

Q9. Which function overcomes the left-lookup limitation of VLOOKUP?

- (A) HLOOKUP (B) XLOOKUP / INDEX-MATCH
(C) SUMIF (D) CHOOSE

Answer

(B) Both XLOOKUP and INDEX-MATCH can return values from any column direction.

Q10. In the formula =INDEX(A1:A10, MATCH("X", B1:B10, 0)), what does MATCH return?

- (A) The value "X" (B) The column number
(C) The row position of "X" in B1:B10 (D) The range B1:B10

Answer

(C) MATCH returns the relative row position of the lookup value within the lookup range.

Q11. Which error appears when a lookup value is not found?

- (A) #REF! (B) #VALUE!
(C) #N/A (D) #NAME?

Answer

(C) #N/A means “Not Available” — the lookup value was not found in the range.

Q12. What does =IFERROR(formula, "Not Found") do?

- (A) Always returns "Not Found" (B) Returns "Not Found" only for #N/A errors
(C) Returns "Not Found" if the formula produces any error (D) Prevents the formula from running

Answer

(C) IFERROR catches any error and returns the specified alternate value.

Q13. In SUMIFS, where does the sum_range argument appear?

- (A) Last (B) Second
(C) Third (D) First

Answer

(D) SUMIFS(sum_range, criteria_range1, criteria1, ...) — the sum range is first, unlike SUMIF where it is last.

Q14. Which formula counts cells in A1:A100 that contain the text "Sales"?

- (A) =COUNT(A1:A100,"Sales") (B) =COUNTIF(A1:A100,"Sales")
(C) =SUMIF(A1:A100,"Sales") (D) =COUNTA(A1:A100,"Sales")

Answer

(B) =COUNTIF(range, criterion) counts cells matching the criterion.

Q15. What does Conditional Formatting allow you to do?

- (A) Run IF formulas automatically (B) Apply visual formatting rules based on cell values or formulas
(C) Create dropdown lists in cells (D) Lock cells from editing

Answer

(B) Conditional Formatting changes cell colour, font, or borders based on rules (value, text, date, or custom formula).

Q16. Which function returns the k -th largest value in a range?

- (A) MAX(range,k) (B) LARGE(range,k)
(C) RANK(range,k) (D) TOP(range,k)

Answer

(B) =LARGE(range, k) returns the k -th largest value. =SMALL(range, k) returns the k -th smallest.

Q17. The DATEDIF function calculates the difference between two dates. Which unit returns the difference in complete months?

- (A) "D" (B) "Y"
(C) "M" (D) "W"

Answer

(C) "M" returns complete months. "D" = days, "Y" = years.

Q18. What is the purpose of Goal Seek in Google Sheets?

- (A) Finds the formula with an error (B) Finds the input value needed to achieve a specific formula result
(C) Creates multiple chart scenarios (D) Automatically fills a series of values

Answer

(B) Goal Seek back-solves: given a desired output, it finds the input that produces it.

Q19. Array formulas in Google Sheets are wrapped in which function?

- (A) ARRAYIF() (B) EXPAND()
(C) ARRAYFORMULA() (D) ARRAY()

Answer

(C) =ARRAYFORMULA() applies a formula across a range and returns an array result.

Q20. Which pivot table area determines how rows are grouped?

- (A) Values (B) Filters
(C) Columns (D) Rows

Answer

(D) The Rows area defines the grouping field displayed on the left side of the pivot table.

Q21. A Slicer in Google Sheets is used for:

- (A) Splitting text to columns (B) Interactive filtering of pivot tables and charts
(C) Removing duplicate values (D) Creating dropdown menus

Answer

(B) Slicers provide clickable buttons to interactively filter one or more pivot tables or charts.

Q22. What does =MOD(10, 3) return?

- (A) 3 (B) 1
(C) 0 (D) 3.33

Answer

(B) MOD returns the remainder: $10 \div 3 = 3$ remainder 1.

Q23. Which function converts a number to a formatted text string (e.g., currency)?

- (A) FORMAT() (B) TEXT()
(C) VALUE() (D) CONVERT()

Answer

(B) =TEXT(value, format) converts a number to text with a specified format: =TEXT(1234.5, "#,##0.00").

Q24. What is the shortcut to open Find & Replace in Google Sheets?

- (A) Ctrl+F (B) Ctrl+H
(C) Ctrl+R (D) Ctrl+G

Answer

(B) Ctrl+H opens Find & Replace. Ctrl+F only opens Find.

Q25. Which formula returns the number of characters in cell A1?

- (A) =COUNT(A1) (B) =SIZE(A1)
(C) =LEN(A1) (D) =LENGTH(A1)

Answer

(C) =LEN(text) returns the number of characters including spaces.

Q26. The WEEKDAY(date, 2) function returns 1 for which day?

- (A) Sunday (B) Saturday
(C) Monday (D) Friday

Answer

(C) With type 2, WEEKDAY returns 1 for Monday and 7 for Sunday.

Q27. In a Scenario Manager, what is stored in each scenario?

- (A) A set of chart formatting options (B) A set of changing cell values and the resulting output
(C) A set of conditional formatting rules (D) A list of formula errors

Answer

(B) Each Scenario saves specific values for the changing cells so you can compare different “what-if” situations.

Q28. Which chart type is best for showing a part-to-whole relationship?

- (A) Line chart (B) Scatter plot
(C) Histogram (D) Pie chart

Answer

(D) Pie charts show how parts contribute to a whole. Use sparingly (max 5–6 slices).

Q29. What does =COUNTA(A1:A10) count?

- (A) Only numeric cells (B) All non-empty cells
(C) Only text cells (D) All empty cells

Answer

(B) COUNTA counts all non-empty cells regardless of data type. COUNT counts only numbers.

Q30. What error does dividing by zero produce?

- (A) #NULL! (B) #N/A
(C) #VALUE! (D) #DIV/0!

Answer

(D) #DIV/0! is produced whenever a formula attempts to divide by zero or an empty cell.

Q31. Which function returns the last day of the month 2 months from a given date?

- (A) EDATE(date,2) (B) EOMONTH(date,2)
(C) ENDMONTH(date,2) (D) MONTHEND(date,2)

Answer

(B) =EOMONTH(date, months) returns the last day of the month that is *n* months away.

Q32. What does =SUBSTITUTE(A1,"old","new") do?

- (A) Finds the position of "old" in A1 (B) Replaces every occurrence of "old" with "new" in A1
(C) Returns TRUE if "old" exists in A1 (D) Counts how many times "old" appears in A1

Answer

(B) SUBSTITUTE(text, old, new) replaces all occurrences of *old* with *new* in the text.

Q33. The #REF! error occurs when:

- (A) A formula references a cell that no longer exists or is invalid
(B) A formula divides by zero
(C) A lookup value is not found
(D) A function name is misspelled

Answer

(A) #REF! means the formula contains a reference to a deleted or invalid cell/range.

Q34. What does the HLOOKUP function search across?

- (A) Columns (vertically)
(B) Rows (horizontally)
(C) A named range
(D) An entire workbook

Answer

(B) HLOOKUP (Horizontal Lookup) searches across the first row of a range and returns a value from a specified row.

Q35. In conditional formatting, a “Custom Formula” rule applies formatting when the formula returns:

- (A) Any numeric value
(B) TRUE
(C) FALSE
(D) An error

Answer

(B) Custom formula rules format the cell when the formula evaluates to TRUE (or any non-zero value).

Q36. Which aggregate function is most resistant to the effect of outliers?

- (A) Mean
(B) Sum
(C) Median
(D) Count

Answer

(C) The Median is resistant to outliers because it is based on position, not magnitude.

2 1**Q37.** What does `np.arange(2, 10, 3)` return?

- (A) `[2, 5, 8]` (B) `[2, 3, 4, 5, 6, 7, 8, 9, 10]`
(C) `[2, 4, 6, 8]` (D) `[3, 6, 9]`

Answer

(A) `np.arange(start, stop, step)` generates `[2, 5, 8]` — starts at 2, stops before 10, step 3.

Q38. What attribute gives the number of dimensions of a NumPy array?

- (A) `.shape` (B) `.size`
(C) `.ndim` (D) `.dims`

Answer

(C) `.ndim` returns the number of dimensions (e.g., 1 for 1D, 2 for 2D).

Q39. Given `a = np.array([[1,2,3],[4,5,6]])`, what is `a[1, 2]`?

- (A) 3 (B) 5
(C) 6 (D) 2

Answer

(C) Row index 1, column index 2 → 6 (second row, third column).

Q40. Which NumPy operation transposes a 2D array A?

- (A) `np.flip(A)` (B) `A.T`
(C) `np.rotate(A)` (D) `A.transpose_all()`

Answer

(B) `A.T` or `np.transpose(A)` swaps rows and columns.

Q41. What does Boolean masking return?

- (A) A boolean scalar (B) A boolean array only
(C) Elements where the mask is TRUE (D) Elements where the mask is FALSE

Answer

(C) Boolean masking filters and returns only the elements of the array where the condition is True.

Q42. Which operator is used for element-wise multiplication of two NumPy arrays?

- (A) `@` (B) `*`
(C) `np.dot()` (D) `**`

Answer

(B) * performs element-wise multiplication. @ or np.dot() performs matrix multiplication.

Q43. What does `np.linspace(0, 1, 5)` produce?

- (A) `[0, 0.25, 0.5, 0.75, 1.0]` (B) `[0, 1, 2, 3, 4]`
(C) `[0, 0.2, 0.4, 0.6, 0.8]` (D) `[0.2, 0.4, 0.6, 0.8, 1.0]`

Answer

(A) `np.linspace(start, stop, num)` returns num evenly spaced values including both endpoints.

Q44. How do you load a CSV file into a Pandas DataFrame?

- (A) `pd.load_csv("f.csv")` (B) `pd.read_csv("f.csv")`
(C) `pd.open("f.csv")` (D) `pd.import_csv("f.csv")`

Answer

(B) `pd.read_csv(filepath)` is the standard function to load a CSV into a DataFrame.

Q45. Which method shows a quick statistical summary of all numeric columns?

- (A) `df.info()` (B) `df.summary()`
(C) `df.describe()` (D) `df.stats()`

Answer

(C) `df.describe()` returns count, mean, std, min, quartiles, and max for numeric columns.

Q46. How do you count missing values per column in a DataFrame `df`?

- (A) `df.count_null()` (B) `df.isnull().sum()`
(C) `df.missing()` (D) `df.nan_count()`

Answer

(B) `df.isnull().sum()` returns the number of NaN values in each column.

Q47. Which method drops rows where the column 'age' is NaN?

- (A) `df.dropna()` (B) `df.drop(columns='age')`
(C) `df.dropna(subset=['age'])` (D) `df.fillna(subset=['age'])`

Answer

(C) `df.dropna(subset=['age'])` drops rows only when the specified column has NaN.

Q48. What does `df.drop_duplicates()` do?

- (A) Drops columns with duplicate names (B) Removes rows that are entirely blank
 (C) Removes duplicate rows, keeping the first occurrence (D) Removes the last row of the DataFrame

Answer

(C) `drop_duplicates()` removes duplicate rows. Default `keep='first'` retains the first occurrence.

Q49. How do you filter a DataFrame to rows where `age > 25` AND `city == 'Delhi'`?

- (A) `df[df['age']>25 and df['city']=='Delhi']` (B) `df[(df['age']>25) & (df['city']=='Delhi')]`
 (C) `df[df['age']>25 & df['city']=='Delhi']` (D) `df.filter(age>25, city='Delhi')`

Answer

(B) Use `&` (bitwise AND) with parentheses around each condition: `(cond1) & (cond2)`.

Q50. What is the difference between `.loc` and `.iloc`?

- (A) `.loc` is faster; `.iloc` is slower (B) `.loc` uses label-based indexing; `.iloc` uses integer-position indexing
 (C) `.loc` works on columns only; `.iloc` on rows only (D) They are identical

Answer

(B) `.loc` = label-based (index names/column names). `.iloc` = integer position (0, 1, 2...).

Q51. What does `df.groupby('dept')['salary'].mean()` compute?

- (A) The mean salary across the entire DataFrame (B) The mean salary for each unique department
 (C) The total salary per department (D) The number of employees per department

Answer

(B) `groupby` splits the data by 'dept' and computes the mean of 'salary' for each group.

Q52. Which `pd.merge()` join type keeps all rows from the left DataFrame?

- (A) inner (B) outer
 (C) right (D) left

Answer

(D) `how='left'` retains all rows from the left DataFrame and fills NaN for non-matching right rows.

Q53. What does `pd.concat([df1, df2], axis=0)` do?

- (A) Merges on a key column
- (B) Stacks df1 and df2 side by side (column-wise)
- (C) Stacks df1 and df2 vertically (row-wise)
- (D) Performs an outer join

Answer

(C) axis=0 stacks DataFrames vertically (adds rows). axis=1 stacks horizontally (adds columns).

Q54. Which function converts a wide DataFrame to long format?

- (A) `df.pivot()`
- (B) `pd.melt()`
- (C) `df.stack()`
- (D) `pd.wide_to_long()`

Answer

(B) `pd.melt()` unpivots a DataFrame from wide to long format (one row per variable-value pair).

Q55. What does `df['col'].astype('category')` do?

- (A) Converts the column to string type
- (B) Converts the column to a boolean
- (C) Converts the column to Pandas category dtype, saving memory
- (D) Converts the column to integer codes

Answer

(C) The category dtype stores repeated string values efficiently by mapping them to integer codes internally.

Q56. How do you compute a 7-day rolling average on column 'sales'?

- (A) `df['sales'].rolling(7).sum()`
- (B) `df['sales'].rolling(window=7).mean()`
- (C) `df['sales'].expanding(7).mean()`
- (D) `df['sales'].window(7).avg()`

Answer

(B) `.rolling(window=7).mean()` computes the moving average over the last 7 periods.

Q57. What is Pearson's correlation coefficient when two variables have no linear relationship?

- (A) 1
- (B) -1
- (C) 0
- (D) 0.5

Answer

(C) A Pearson's $r = 0$ indicates no linear relationship between the two variables.

Q58. Which Seaborn function creates a matrix of pairwise scatter plots?

- (A) `sns.heatmap()` (B) `sns.scatterplot()`
(C) `sns.pairplot()` (D) `sns.jointplot()`

Answer

(C) `sns.pairplot(df)` plots all pairwise bivariate scatter plots and univariate histograms.

Q59. In the IQR outlier method, the lower fence is:

- (A) $Q1 - 2 \times IQR$ (B) $Q1 - 1.5 \times IQR$
(C) $Q1 - IQR$ (D) $Q2 - 1.5 \times IQR$

Answer

(B) Lower fence = $Q1 - 1.5 \times IQR$. Upper fence = $Q3 + 1.5 \times IQR$.

Q60. Z-score outliers are typically defined as observations where $|z|$ is greater than:

- (A) 1 (B) 2
(C) 3 (D) 5

Answer

(C) Values with $|z| > 3$ are typically flagged as outliers, as they lie more than 3 standard deviations from the mean.

Q61. What does `df['sales'].pct_change()` compute?

- (A) Cumulative sum (B) Percentage of total
(C) Period-over-period percentage change (D) Rolling percentage change

Answer

(C) `pct_change()` computes $(x_t - x_{t-1})/x_{t-1}$ — the relative change from one period to the next.

Q62. Skewness of 0 means the distribution is:

- (A) Right-skewed (B) Left-skewed
(C) Symmetric (normal-like) (D) Bimodal

Answer

(C) Skewness = 0 indicates a symmetric distribution. Positive = right-skewed; negative = left-skewed.

Q63. Which plot is best for visualising the distribution of a single continuous variable?

- (A) Scatter plot (B) Bar chart
(C) Histogram / KDE plot (D) Line chart

Answer

(C) A histogram (or KDE plot) shows the distribution of a single continuous variable.

Q64. What is the median of the values [3, 7, 1, 9, 5]?

- (A) 3 (B) 7
(C) 5 (D) 9

Answer

(C) Sorted: [1, 3, 5, 7, 9] → median = 5 (middle value of 5 elements).

Q65. Which Pandas function creates a pivot table with aggregation?

- (A) `df.pivot()` (B) `df.melt()`
(C) `pd.pivot_table()` (D) `df.crosstab()`

Answer

(C) `pd.pivot_table()` supports `aggfunc` and `fill_value`; `df.pivot()` does not aggregate.

Q66. In a bivariate analysis, `sns.regplot()` adds:

- (A) A KDE curve (B) A regression line with confidence interval
(C) A second Y-axis (D) Error bars

Answer

(B) `sns.regplot()` plots a scatter plot with a linear regression line and a shaded confidence interval band.

Q67. What does `df.query("age > 25 and city == 'Mumbai'")` return?

- (A) A boolean Series (B) Rows where age > 25 and city is Mumbai
(C) Rows where age > 25 or city is Mumbai (D) A count of matching rows

Answer

(B) `df.query()` filters rows matching the string expression, equivalent to boolean indexing.

Q68. Which NumPy function computes the matrix inverse?

- (A) `np.inv()` (B) `np.linalg.inv()`
(C) `np.matrix.inverse()` (D) `np.solve()`

Answer

(B) `np.linalg.inv(A)` computes the inverse of matrix A. The matrix must be square and non-singular.

Q69. What does `df['col'].expanding().sum()` compute?

- (A) Rolling sum over last N periods
- (B) Cumulative sum from the first row to the current row
- (C) Sum of the entire column
- (D) Percentage contribution of each row

Answer

(B) `expanding()` grows the window from the start; `.sum()` gives the cumulative sum at each point.

Q70. To compute the frequency table of a categorical column, use:

- (A) `df['col'].count()`
- (B) `df['col'].value_counts()`
- (C) `df['col'].frequency()`
- (D) `df['col'].groupby().count()`

Answer

(B) `value_counts()` returns a Series of counts for each unique value, sorted descending.

Q71. A positive skewness value means:

- (A) Mean < Median (left tail is longer)
- (B) Mean = Median (symmetric)
- (C) Mean > Median (right tail is longer)
- (D) The distribution has two peaks

Answer

(C) Positive skewness = right-skewed distribution = long tail on the right = Mean > Median > Mode.

Q72. Which seaborn plot shows distribution AND individual data points?

- (A) `sns.boxplot()`
- (B) `sns.violinplot()`
- (C) `sns.stripplot()`
- (D) `sns.violinplot()` combined with `sns.stripplot()`

Answer

(D) Combining violin + strip plot shows the full distribution shape and individual data points together.

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Q73. In Tableau, a **blue pill** on a shelf represents:

- (A) A continuous field (creates an axis) (B) A calculated field
(C) A discrete field (creates a header) (D) A geographic dimension

Answer

(C) Blue = Discrete → creates headers/labels. Green = Continuous → creates a numeric axis.

Q74. What is the difference between a Live connection and an Extract in Tableau?

- (A) Live is faster; Extract is slower (B) Live queries data in real-time; Extract is a saved snapshot (.hyper) for faster performance
(C) Extract is always more accurate than Live (D) They are the same; just different names

Answer

(B) Live = real-time queries. Extract = saved local snapshot (.hyper file), faster but requires scheduled refresh.

Q75. Which Tableau feature is recommended for combining tables from the SAME data source?

- (A) Data Blending (B) Join / Relationship
(C) Union (D) Cross-database join

Answer

(B) Relationships (default since v2020.2) and Joins both work within the same source. Relationships are the recommended default.

Q76. What is the purpose of a Context Filter in Tableau?

- (A) Filters data before it is visualised (B) Creates a temporary filtered dataset that other filters run against, improving performance
(C) Removes null values from the view (D) Applies a filter across all dashboards

Answer

(B) A Context Filter creates a temporary “context” table; all subsequent filters operate on this smaller set, improving speed.

Q77. What does a **FIXED LOD** expression do?

- | | |
|--|--|
| (A) Computes at the view level of detail | (B) Adds extra dimensions to the view |
| (C) Computes at a specified dimension level, ignoring view-level dimension filters | (D) Excludes specified dimensions from the computation |

(C) **FIXED** locks the aggregation level to the specified dimension(s) and ignores dimension filters (but respects context filters).

Q78. Which LOD expression ADDS dimensions to the view's level of detail?

- (A) FIXED
 - (B) EXCLUDE
 - (C) INCLUDE
 - (D) WINDOW

(C) INCLUDE adds extra dimensions to the existing view granularity, making the calculation more granular.

Q79. Which LOD expression REMOVES dimensions from the view's level of detail?

- (A) FIXED (B) EXCLUDE
(C) INCLUDE (D) DETAIL

(B) **EXCLUDE** removes specified dimensions from the view’s level of detail, making the aggregation less granular.

Q80. What is the correct Tableau LOD syntax for total sales per customer?

- | | | | |
|-------------------|--------------------------|----------------|--------------------------------|
| (A) | {INCLUDE [Sales] : | (B) | {FIXED [Customer ID] : |
| SUM([Customer]))} | | SUM([Sales]))} | |
| (C) | {EXCLUDE [Customer ID] : | (D) | SUM({[Customer ID] : [Sales]}) |
| SUM([Sales]))} | | | |

Answer

(B) $\{\text{FIXED [Customer ID] : SUM([Sales])}\}$ computes total sales per customer regardless of what is in the view.

Q81. A Tableau Parameter differs from a filter because:

- (A) A Parameter only works with dates (B) A Parameter is a user-controlled variable that can replace constants in calculations
- (C) A Parameter filters data at the source level (D) A Parameter can only hold integer values

Answer

(B) A Parameter is a dynamic value set by the user; it can be used in calculated fields, reference lines, and filters — not just to filter rows.

Q82. A Set in Tableau defines:

- (A) A group of calculated fields
- (B) A custom subset of dimension members
- (C) A type of chart
- (D) A dashboard action

Answer

(B) A Set is a custom field that marks dimension members as IN or OUT based on a condition or manual selection.

Q83. What is the difference between a Fixed Set and a Dynamic Set in Tableau?

- (A) Fixed sets update with data; Dynamic sets are static
- (B) Fixed sets are manually selected members; Dynamic sets are condition-based and update with data
- (C) They are the same thing with different names
- (D) Fixed sets are for measures; Dynamic sets are for dimensions

Answer

(B) Fixed sets = manually chosen members (static). Dynamic sets = based on a condition, update when data refreshes.

Q84. Which Tableau feature allows users to drill down through a date hierarchy (Year → Quarter → Month)?

- (A) Bins
- (B) Groups
- (C) Hierarchies
- (D) Sets

Answer

(C) Hierarchies allow users to expand (+) and collapse (-) levels to drill down or roll up.

Q85. How are Bins in Tableau typically used?

- (A) To group dimension members manually
- (B) To bucket continuous numeric measures into ranges for histograms
- (C) To create calculated fields
- (D) To define context filters

Answer

(B) Bins divide a continuous measure into equal-width buckets, which can then be used like a dimension to build histograms.

Q86. What does a Table Calculation in Tableau operate on?

- (A) The underlying database rows
- (B) The aggregated values already returned in the view
- (C) The extract file directly
- (D) The data source before filters

Answer

(B) Table Calculations work on the aggregated data already in the view — they are applied after the query result is returned.

Q87. Which Table Calculation returns a cumulative total?

- (A) RANK() (B) LOOKUP()
(C) RUNNING_SUM() (D) WINDOW_AVG()

Answer

(C) RUNNING_SUM(SUM([Sales])) computes the cumulative (running) total from the first mark to the current mark.

Q88. What is the correct filter order in Tableau from first to last applied?

- (A) Dimension → Measure → Context → Extract
(B) Extract → Data Source → Context → Dimension → Measure → Table Calculation
(C) Table Calc → Measure → Dimension → Context → Extract
(D) Context → Extract → Dimension → Measure → Table Calc

Answer

(B) Tableau filter order: Extract → Data Source → Context → Sets → Dimension → Measure → Table Calculation.

Q89. In Tableau, Data Blending is used when:

- (A) Two tables are in the same data source
(B) You want to stack tables with the same schema
(C) You want to combine data from different data sources
(D) You want to perform a cross join

Answer

(C) Data Blending links fields from a primary and secondary source that come from different connections.

Q90. What is a Union in Tableau?

- (A) Combines tables with a matching key column
(B) Stacks tables with the same structure vertically (like SQL UNION ALL)
(C) Merges fields from two different data sources
(D) Creates a Cartesian product of two tables

Answer

(B) A Union appends rows from multiple tables that have the same columns/structure, stacking them vertically.

Q91. Which Marks Card property disaggregates data without changing the visual encoding?

- (A) Color (B) Size
(C) Label (D) Detail

Answer

(D) Detail adds a dimension to the level of detail of the mark without affecting colour, size, or shape.

Q92. In a Tableau Dashboard, what is the difference between Tiled and Floating objects?

- (A) Tiled objects are charts; Floating objects are filters
(B) Tiled objects snap to the layout grid; Floating objects can be positioned freely anywhere
(C) Floating objects are part of containers; Tiled objects are not
(D) They behave identically

Answer

(B) Tiled = snaps to the grid within containers. Floating = placed freely at any position and can overlap other objects.

Q93. Which dashboard action filters other sheets when a user clicks a mark?

- (A) URL Action (B) Highlight Action
(C) Filter Action (D) Parameter Action

Answer

(C) A Filter Action uses the selected mark's field values to filter the data shown in target sheets.

Q94. What is the purpose of a Highlight Action in Tableau?

- (A) It filters out non-selected marks (B) It opens a URL based on a field value
(C) It emphasises related marks in other sheets without removing the others (D) It changes the colour encoding of a chart

Answer

(C) A Highlight Action visually emphasises matching marks in target sheets while keeping all other marks visible (greyed out).

Q95. A Tableau Story is best described as:

- (A) A type of calculated field (B) A sequence of worksheets/dashboards arranged as story points to convey a narrative
(C) A data source with multiple tables (D) A type of LOD expression

Answer

(B) A Story in Tableau is a presentation-style feature with ordered story points, each showing a view with a caption.

Q96. Which layout container stacks objects left to right in Tableau?

- (A) Vertical Container
- (B) Tiled Container
- (C) Horizontal Container
- (D) Flow Container

Answer

(C) A Horizontal Container arranges child objects side by side and auto-adjusts widths.

Q97. Which of the following is a Tableau Relationship characteristic?

- (A) Performs a physical join at the row level before analysis
- (B) Requires tables to have the same granularity
- (C) Defers join type to query time, preserving each table's granularity
- (D) Can only be used with Extract connections

Answer

(C) Relationships are flexible — the join type (inner/left) is determined at query time based on which fields are used, preserving row granularity.

Q98. What does placing a **continuous** date field on Rows create?

- (A) Category headers for each year
- (B) A continuous date axis
- (C) Discrete labels for each month
- (D) A hierarchy with drill-down

Answer

(B) A continuous (green) date creates a continuous axis, while a discrete (blue) date creates headers.

Q99. Tableau Groups differ from Sets because:

- (A) Groups are dynamic; Sets are static
- (B) Groups manually combine dimension members into named categories; Sets classify members as In or Out
- (C) Groups work on measures; Sets work on dimensions
- (D) They are identical

Answer

(B) Groups merge specific members into a named bucket (e.g., "East + West = Coast"). Sets classify members as In/Out based on conditions.

Q100. Which calculated field function returns a part of a date (e.g., the year)?

- (A) DATEYEAR()
- (B) DATEPART('year', [Date])
- (C) YEAR_PART([Date])
- (D) DATE_FORMAT([Date], 'yyyy')

Answer

(B) DATEPART('year', [Date]) returns the numeric year of the date. Other options are not valid Tableau functions.

Q101. What is “Viz in Tooltip” in Tableau?

- (A) A tooltip that opens a new worksheet
- (B) A small embedded visualisation that appears in the hover tooltip of a mark
- (C) A filter applied via tooltip
- (D) A chart type for mobile dashboards

Answer

(B) Viz in Tooltip embeds a mini-chart inside the tooltip that appears when a user hovers over a mark, giving extra context.

Q102. Which type of Tableau filter ignores dimension filters applied in the view?

- (A) Measure Filter
- (B) Data Source Filter
- (C) Table Calculation Filter
- (D) FIXED LOD (used as a filter)

Answer

(D) A FIXED LOD expression used as a dimension filter runs before regular dimension filters, so it effectively ignores them (unless a context filter is applied).

Q103. In Tableau, what does “Show Me” do?

- (A) Displays the underlying data table
- (B) Suggests appropriate chart types based on the fields currently selected
- (C) Opens the calculated field editor
- (D) Runs a query against the data source

Answer

(B) Show Me recommends and switches between chart types based on the number and type of selected fields (dimensions vs measures).